



Rajarata University of Sri Lanka

Faculty of Applied Sciences

Department of Information Technology

ICT2403 - Graphics and Image Processing

Submission 02 - Video Acquisition Method

Submitted by:-

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1. Video Capture Device

Video footage for this project was captured using a mobile phone (Apple iPhone 11) equipped with a rear-facing camera. The mobile phone camera was selected due to its wide availability, portability, and ability to record high-quality video suitable for road surface inspection tasks. Modern mobile phone cameras provide sufficient spatial resolution, and stable recording capabilities, making them appropriate for capturing detailed road surface conditions without the need for specialized equipment.

2. Recording Parameters and their Significance

a) Resolution

All videos were recorded at a resolution of 1920 x 1080 pixels (full HD). This resolution was chosen to ensure that fine road surface details such as cracks, pothole edges, and surface texture variations are clearly visible in individual frames. A higher resolution improves the accuracy of segmentation while maintaining a reasonable balance between visual detail and computational efficiency during frame processing.

b) Frame Rate

A frame rate of 30 FPS was used for video recording. This frame rate provides smooth motion continuity and minimizes motion blur caused by camera movement. Maintaining a sufficient frame rate is important for accurate frame extraction and analysis, particularly when recording videos with handheld motion or slow vehicle movement.

c) Duration of Video

Each video clip had a duration between 15 to 30 seconds. This duration was sufficient to cover the selected road segments while providing an enough number of frames for image enhancement, segmentation, and evaluation. Short video durations also reduce redundancy and simplify data annotation and processing.

3. Environmental Condition

Video recording was carried out under varying environmental conditions during data collection process. At the beginning of the recording session, videos were captured during a bright sunny day under natural daylight conditions, which provided sufficient illumination of the road surface. During the later part of the recording session, the last two to three video clips were recorded under rainy conditions. As a result, the road surface appeared wet in some frames, with reflections and brightness variations.

Additionally, shadows caused by our team members' umbrellas were present in certain video clips due to strong sunlight conditions, even during periods of rainfall. These lighting variations and shadow effects represent realistic environmental challenges commonly happens in real-world road monitoring scenarios and were considered during the following image enhancement stage of the project.

4. Camera Motion and Placement

The camera was positioned using both downward-facing and slightly angled orientations depending on the road segment and surrounding conditions. A downward-facing orientation was primarily used to focus directly on the road surface, while a slight forward angle was adopted in some recordings to capture wider view of the damaged regions ahead. The approximate camera height from the road surface was between 0.3 and 1.0 meters.

Most videos were recorded using handheld motion at a slow walking speed to reduce unnecessary motion blur and sudden vibrations. In addition, a small number of video clips were recorded by one team member while travelling on a slow-moving vehicle within residential areas near their home. Vehicle-based recordings were performed at low speed to maintain video stability and ensure clear visibility of road surface conditions. This combination of handheld and vehicle-mounted recording methods allowed the dataset to capture diverse real-world recording scenarios.

5. Site Selection

Road segments were selected from the Mihintale, Kanadarawa and Anuradhapura New Town area, as well as from residential roads near the group members' homes, covering both tar roads and concrete roads. These locations were chosen due to the presence of clearly visible road surface damages commonly observed in urban, semi-urban and rural environments. The recorded damage types include potholes, cracks, patch failures, edge cracking and road marking removal.

Priority was given to road sections with moderate or low traffic to ensure safe recording conditions and minimal visual obstruction. Including road segments from different locations and residential areas allowed the dataset to capture different road surface conditions and damage patterns, making it suitable for developing and evaluating a generalized video-based road damage detection system.

6. Recording Process and Sample Videos

Multiple video clips were recorded from each selected road segment to capture different damage types and surface conditions. A total of several short video clips were collected during the recording process. Videos that contained unnecessary motion blur, obstruction by vehicle or people or unstable camera movement were deleted. Only visually clear and stable recordings were saved for further analysis and dataset preparation.

7. Video Analysis and Dataset Preparation

All recorded videos were stored in MP4 format. After recording, videos were reviewed to identify suitable segments for frame extraction. Frames were extracted from the videos using a **Uniform Sampling Strategy**, where every n th (10th) frame was selected to reduce redundancy while maintaining representative coverage of the road surface. The extracted frames were resized where necessary and organized into structures folders based on road type and damage category to facilitate further processing.

8. Data Annotation Process

Manual annotation was performed on selected frames to create ground truth data for evaluation purposes. In the current stage of the project, **bounding box annotation** was applied to mark road surface damage regions, and each annotated region was labeled according to the corresponding damage type such as potholes, cracks, patch failures, edge cracking and marking removal. Bounding box annotation was carried out using the **Labelling** annotation tool, which provides an efficient and structured approach for labeling damage regions. The annotated dataset act as reliable reference data for validating the segmentation performance of the proposed system in following stages of the project.

9. Ethical and Safety Considerations

Ethical and safety guidelines were strictly followed during video data collection. Recordings were conducted without interfering with traffic or road users. No identifiable individuals, vehicle license plates, or sensitive information were intentionally captured in the videos. All recordings were performed in publicly accessible areas, ensuring compliance with ethical standards and responsible data collection practices.